



University of Information Technology and Sciences

Software Engineering and System Analysis Lab CSE 356

Software Requirements Specification (SRS) for TouRz

A Smart Online Tour & Travel Management System

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Revision History

Version	Date	Description	Author
1.0	April 2026	Initial SRS Creation	Project Team

1. Introduction

1.1 Purpose

This SRS defines all functional and non-functional requirements for **TourZ** — a web-based tour booking and travel management platform. It is the authoritative reference for development, testing, and validation.

1.2 Document Conventions

Convention	Meaning
REQ-XX	Functional requirement ID
SHALL	Mandatory
High/Medium/Low	Priority for v1.0

1.3 Intended Audience and Reading Suggestions

AUDIENCE	RELEVANT SECTIONS
Developers	All, especially 3, 4, 6
Project Managers	1, 2, 5
Testers / QA	4, 5
Instructors / Evaluators	1, 2, 4, Appendices
End Users	2.2, 2.3, 4

1.4 Product Scope

TourZ is a full-stack web application targeting travelers and agencies in Bangladesh. It eliminates manual booking errors, absent automated ticketing, and lack of 24/7 support by providing: - End-to-end booking: register → browse → book → pay → receive ticket - Automated QR code and PDF ticket generation - AI chatbot for 24/7 support - Admin panel for full platform control

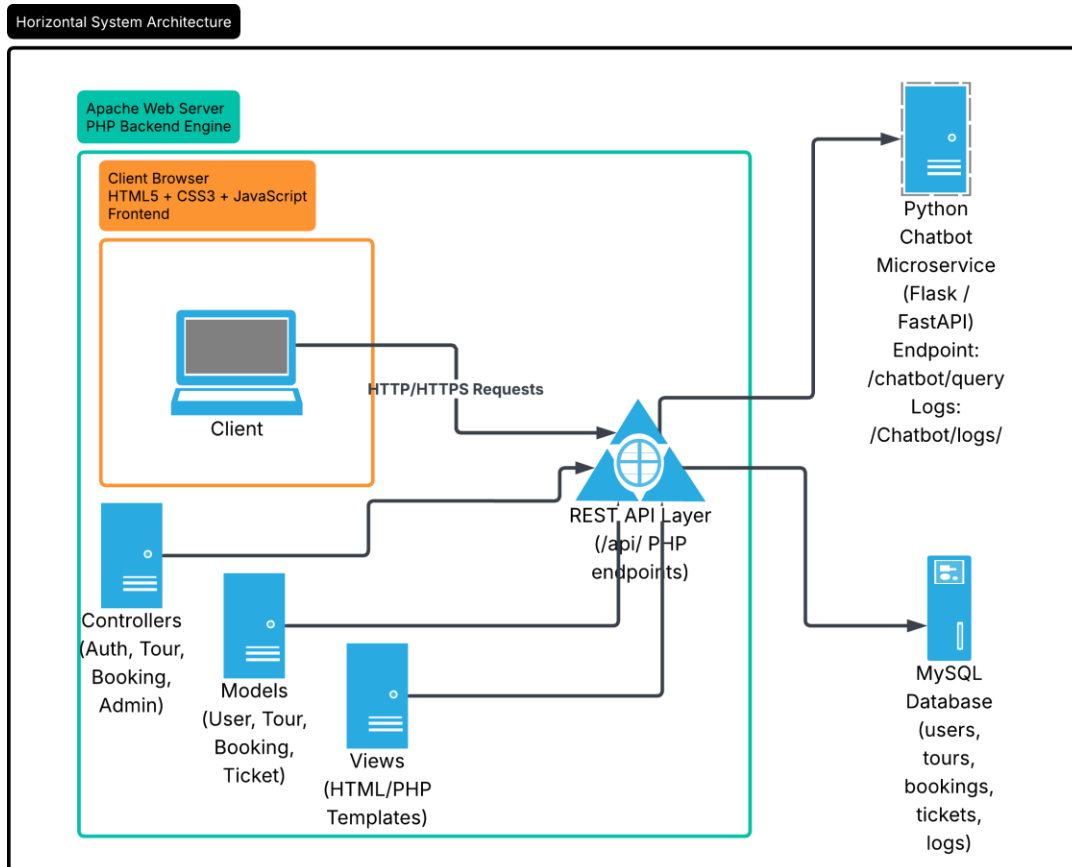
1.5 References

1. Pressman, R. S. *Software Engineering*, 7th Ed.
2. PHP – <https://www.php.net> | MySQL – <https://dev.mysql.com/doc>
3. Flask – <https://flask.palletsprojects.com> | phpqrcode – <http://phpqrcode.sourceforge.net>

2. Overall Description

2.1 Product Perspective

TourZ is a self-contained MVC web application:



2.2 Product Functions

Customer: Register/login · browse/filter tours · book · mock payment · QR/PDF ticket · custom requests · chatbot · profile management.

Admin: Tour CRUD · manage users · view/cancel bookings · review custom requests · chatbot logs · reports.

2.3 User Classes and Characteristics

User Class	Access Level	Priority
Guest	Browse tours only	Low
Registered Customer	Full customer features	High (Primary)

User Class	Access Level	Priority
Administrator	Full system access	Critical

2.4 Operating Environment

Component	Specification
Web Server	Apache 2.4.x (XAMPP / Linux LAMP)
Backend	PHP 8.x
Database	MySQL 8.x
Chatbot	Python 3.10+, Flask 2.x
Browsers	Chrome/Firefox/Edge 100+, Safari 15+
Responsive	320px (mobile) → 1920px+ (desktop)

2.5 Design and Implementation Constraints

1. Backend: **PHP**; Chatbot: **Python (Flask)**; DB: **MySQL** — mandatory.
 2. Frontend: **HTML5 + CSS3 + JS** — no heavy frameworks.
 3. QR via **phpqrcode**; PDFs via **FPDF/TCPDF**.
 4. PHP ↔ Python: **RESTful JSON** only.
 5. Schema: minimum **3NF**; passwords: **bcrypt**; queries: **PDO prepared statements**; **CSRF** on all forms.
 6. PHP session-based auth; admin accounts database-seeded only.
 7. Payment: **mock in v1.0**; architecture must allow real gateway replacement.
-

2.6 User Documentation

Document	Format
User Manual (registration, booking, ticket download)	PDF / Online
Admin Manual (tour, user, booking management)	PDF / Online
Installation Guide (XAMPP / Apache)	Markdown
API Documentation	Markdown
Database Schema Reference (ERD)	PDF / Image

2.7 Assumptions and Dependencies

Users have modern browsers. Server runs XAMPP/LAMP. Python chatbot on port 5000. Payment is mock in v1.0.

Dependency	Purpose
PHP 8.x, MySQL 8.x, Python 3.10+	Core runtimes
Flask / flask-cors	Chatbot HTTP server
phpqrcode, FPDF/TCPDF	QR + PDF generation
Apache HTTP Server	Serve PHP application

3. External Interface Requirements

3.1 User Interfaces

- Fully responsive (320px → 1920px+); dark mode default (localStorage persisted).
- Consistent navbar, footer, inline red errors, green success banners.

Page	Description
Landing Page	Hero CTA, featured tours, testimonials
Tour Listings	Filterable card grid (Domestic / International)
Tour Detail	Description, gallery, pricing, itinerary, Book Now
Registration / Login	Auth forms with client-side validation
User Dashboard	Profile, booking history, custom request status
Booking Flow	Step 1: Details → Step 2: Payment → Step 3: QR + PDF
Chatbot Widget	Floating bubble on all pages
Admin Dashboard	Stats cards, charts, quick actions
Admin Panels	Tour/User/Booking/Request/Log CRUD views

3.2 Hardware Interfaces

Web application — no direct hardware interface. Server minimum: dual-core 1.8 GHz, 2 GB RAM, 4 GB storage. Clients: any modern device.

3.3 Software Interfaces

Component	Version	Interface	Purpose
MySQL	8.x	PHP PDO	Data storage
PHP	8.x	Native	Backend logic
Python/Flask	3.10+/2.x	REST/JSON	Chatbot
phpqrcode	Latest	PHP include	QR generation

Component	Version	Interface	Purpose
FPDF/TCPDF	Latest	PHP include	PDF generation
Apache	2.4.x	HTTP	Serve PHP

3.4 Communications Interfaces

- HTTP (dev) / HTTPS TLS 1.2+ (production).
 - Chatbot: POST `http://localhost:5000/chatbot/query` — `{"query":"..."}` → `{"response":"..."}`.
 - PHP–MySQL: Port 3306 via PDO. Production: secure HTTP-only session cookies.
 - Email (TBD-01): Future SMTP via PHPMailer / SendGrid.
-

4. System Features

4.1 User Registration and Authentication

Priority: High — Secure account creation, login, and logout.

- **REQ-01:** Registration collects Full Name, Email, Password, Phone.
 - **REQ-02:** Validate email format; password ≥ 8 chars with at least one number.
 - **REQ-03:** Hash passwords with `password_hash()` / BCrypt.
 - **REQ-04:** Reject duplicate emails; create PHP session on login storing `user_id` and `role`.
 - **REQ-05:** Redirect unauthenticated access to login; enforce user/admin roles on all routes.
-

4.2 User Profile Management

Priority: Medium — View and update profile and booking history.

- **REQ-06:** Display and allow update of name, phone, and email (if not already taken).
 - **REQ-07:** Booking history shows: destination, date, team size, price, status.
-

4.3 Tour Browsing and Search

Priority: High — All visitors browse and filter tour packages.

- **REQ-08:** Display all tours from `tours` table; cards show name, image, category, price/person.

- **REQ-09:** Category filter (Domestic / International); detail page with description, gallery, itinerary.
 - **REQ-10:** “Book Now” redirects guests to login, authenticated users to booking flow.
-

4.4 Tour Booking System

Priority: High — 3-step booking flow for authenticated customers.

- **REQ-11:** 3-step flow: Details → Payment → Confirmation.
 - **REQ-12:** Collect: Name, Email, Phone, Destination (pre-filled), Team Size; auto-calculate $\text{total_price} = \text{price_per_person} \times \text{team_size}$.
 - **REQ-13:** Save booking to bookings table linked to `user_id`; assign unique Booking ID.
 - **REQ-14:** Confirmation shows Booking ID, destination, team size, price, QR code.
-

4.5 Payment Processing

Priority: High (Mock in v1.0) — Simulated payment; modular for future gateway replacement.

- **REQ-15:** Show booking summary on payment page; mock form with method selector.
 - **REQ-16:** Simulate successful payment and advance to confirmation. Payment module must be replaceable without affecting other modules.
-

4.6 QR Code Ticket Generation

Priority: High — Unique QR code per confirmed booking.

- **REQ-17:** Generate QR per booking using `phpqrcode`; encode Booking ID.
 - **REQ-18:** Save PNG to `/Storage/qr_codes/qr_<booking_id>.png`; store path in DB; display on confirmation page.
-

4.7 PDF Ticket Generation

Priority: High — Downloadable PDF ticket per confirmed booking.

- **REQ-19:** Generate PDF using `FPDF/TCPDF`; include TourZ header, Booking ID, name, destination, date, team size, price, QR image.
- **REQ-20:** Save to `/Storage/pdf_tickets/ticket_<booking_id>.pdf`; provide Download Ticket button on confirmation and booking history.

4.8 Custom Tour Request System

Priority: Medium — Users request destinations not on the platform.

- **REQ-21:** Form collects: Destination, Preferred Dates, Group Size, Notes; save with `status = pending`.
 - **REQ-22:** Admin panel lists all pending requests with Approve / Reject actions.
-

4.9 AI-Based Chatbot Support

Priority: Medium — 24/7 rule-based chatbot via Python Flask microservice.

- **REQ-23:** Floating chat widget on all customer-facing pages.
 - **REQ-24:** PHP proxies query to `POST /chatbot/query`; chatbot handles booking, pricing, and support queries. Logs all interactions to `/Chatbot/logs/chatbot_queries.log`.
 - **REQ-25:** Admin Chatbot Logs viewer shows timestamp, query, and response.
-

4.10 Admin Panel

Priority: High — Secure dashboard for full platform management.

- **REQ-26:** Admin panel restricted to admin role; dashboard shows total users, bookings, and revenue.
 - **REQ-27:** Tour CRUD: add (Name, Description, Category, Price/Person, Image, Link), edit, delete (with confirmation).
 - **REQ-28:** Paginated user list (Name, Email, Phone, Date); view all bookings with filters; cancel bookings.
-

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Page render $\leq$ **3 sec** ($\leq$ 50 users); chatbot $\leq$ **2 sec**; QR $\leq$ **1 sec**; PDF $\leq$ **3 sec**.
 - DB supports $\geq$ 10,000 users and $\geq$ 50,000 bookings without degradation.
 - Use indexed foreign keys and PDO prepared statements; cache static assets.
-

5.2 Safety Requirements

- DB failures $\rightarrow$ user-friendly error, no raw PHP traces.

- Chatbot unavailable → “Support temporarily unavailable”, no crash.
- QR/PDF failure SHALL NOT prevent booking from saving.
- Daily DB backups; critical errors logged to /Storage/logs/.

5.3 Security Requirements

ID	Requirement
REQ-SEC-01	Passwords hashed with BCrypt — never stored plaintext
REQ-SEC-02	All DB queries via PDO prepared statements (prevent SQLi)
REQ-SEC-03	CSRF tokens on all HTML forms
REQ-SEC-04	All input sanitized; all HTML output escaped via htmlspecialchars()
REQ-SEC-05	Secure HTTP-only cookies; session ID regenerated on login
REQ-SEC-06	Role-based access control on every protected route
REQ-SEC-07	HTTPS-ready for production; file uploads validated (JPEG/PNG/WebP, max 5 MB)

5.4 Software Quality Attributes

Attribute	Requirement
Usability	First-time user completes booking unaided; key actions ≤ 3 clicks
Reliability	99% uptime; core features independent of chatbot
Maintainability	MVC separation; non-trivial functions commented
Scalability	Supports new categories, roles, integrations without redesign
Portability	Runs on any Apache + PHP + MySQL (XAMPP / LAMP)
Testability	Each module independently unit-testable

5.5 Business Rules

- Customers must be logged in to book; guests may browse.
- Confirmed bookings are customer-immutable — only admin may cancel.
- Only admin role manages tours and users; accounts are DB-seeded only.
- Price is admin-defined; dynamic pricing not supported in v1.0.
- Chatbot logs all queries; custom requests stay pending until admin acts.

6. Other Requirements

6.1 Database Requirements

Table	Key Columns
users	id, name, email, password, phone, role
tours	id, name, description, category, price_per_person, image, link
bookings	id, user_id, tour_id, team_size, total_price, status, booked_at
tickets	id, booking_id, qr_path, pdf_path, generated_at
custom_requests	id, user_id, destination, preferred_dates, group_size, status
chatbot_logs	id, user_id, query_text, response_text, queried_at

Schema MUST follow 3NF.

6.2 Internationalization Requirements

English only in v1.0. UI strings SHOULD be centralized for future Bengali localization.

6.3 Legal Requirements

Footer SHALL link to Terms & Conditions and Privacy Policy (TBD-06). User data stored only as necessary; not shared with third parties.

6.4 Reuse Objectives

Auth middleware reusable across user/admin routes. QR and PDF functions encapsulated as standalone PHP utility classes.

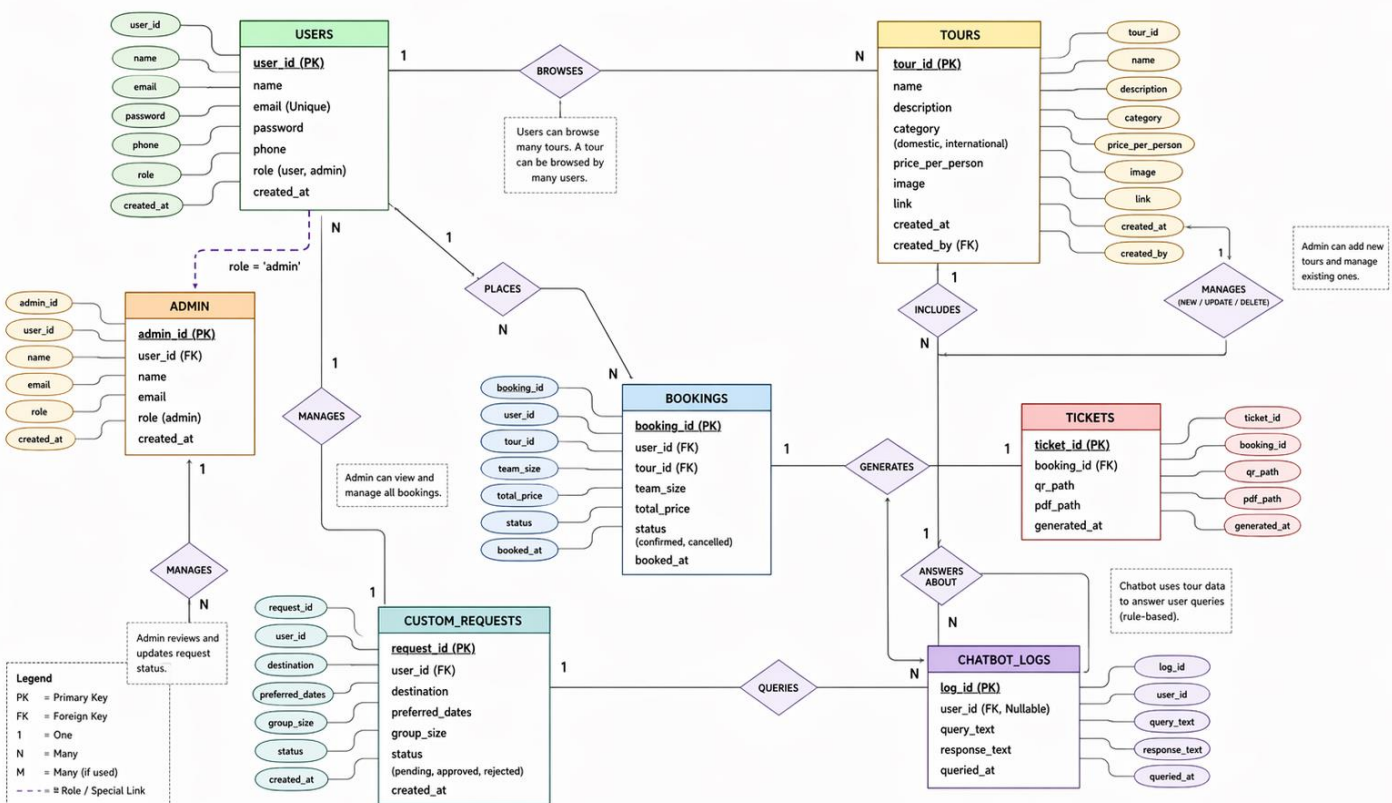
Appendix A: Glossary

Term	Definition
bcrypt	Password hashing algorithm via <code>password_hash()</code>
Booking	Confirmed tour reservation by a registered user
CORS	Cross-Origin Resource Sharing HTTP header mechanism
CSRF	Cross-Site Request Forgery form token protection
ERD	Entity Relationship Diagram
FPDF/TCPDF	PHP libraries for PDF generation
MVC	Model-View-Controller architectural pattern
phpqrcode	PHP library for QR code image generation

Term	Definition
QR Code	Quick Response Code encoding Booking ID
REST	Representational State Transfer API style
SRS	Software Requirements Specification — this document
TBD	To Be Determined
XAMPP	Apache + MySQL + PHP local development bundle
XSS	Cross-Site Scripting attack vector

Appendix B: Analysis Models

B.1 Entity Relationship Diagram

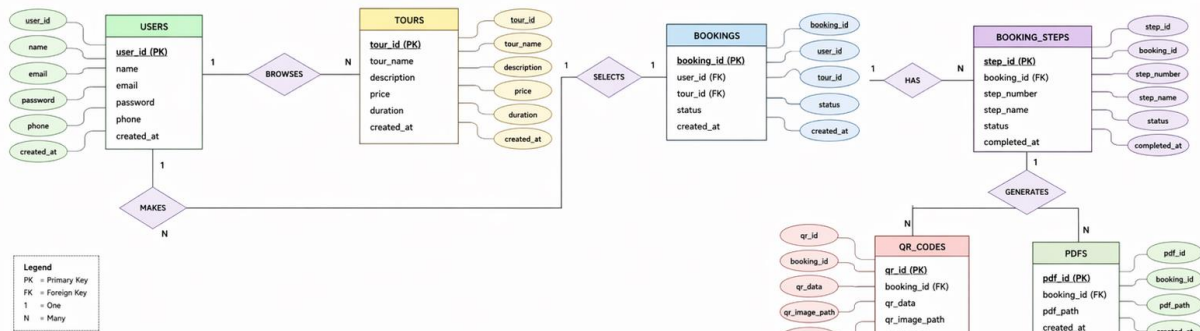


B.2 Use Case Summary

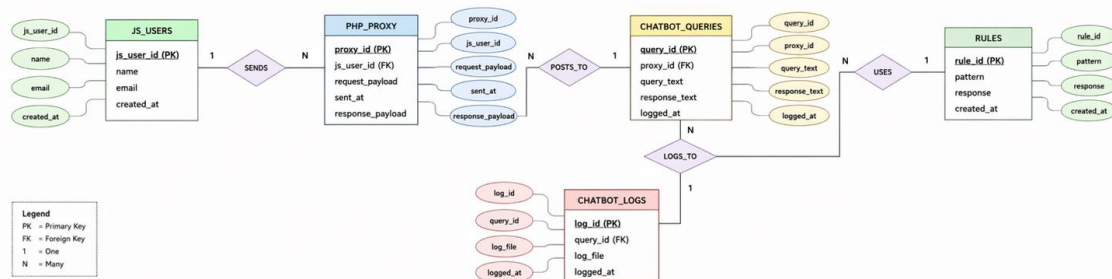
Actor	Use Cases
Guest	Browse tours, view details, access login/register
Customer	Register, login, book, payment, QR/PDF ticket, custom request, profile, chatbot

Actor	Use Cases
Admin	Tour/User/Booking/Request management, chatbot logs, reports

B.3 Booking Flow



B.4 Chatbot Flow



Appendix C: To Be Determined List

ID	Section	Item
TBD-01	3.4	SMTP provider (PHPMailer / SendGrid)
TBD-02	4.5	Real payment gateway (bKash / Nagad / Stripe)
TBD-03	4.1	Forgot Password / email reset flow
TBD-04	5.1	Formal load testing benchmarks
TBD-05	4.9	Chatbot NLP upgrade (spaCy / OpenAI)
TBD-06	2.4	Production hosting provider
TBD-07	4.7	Final PDF ticket template